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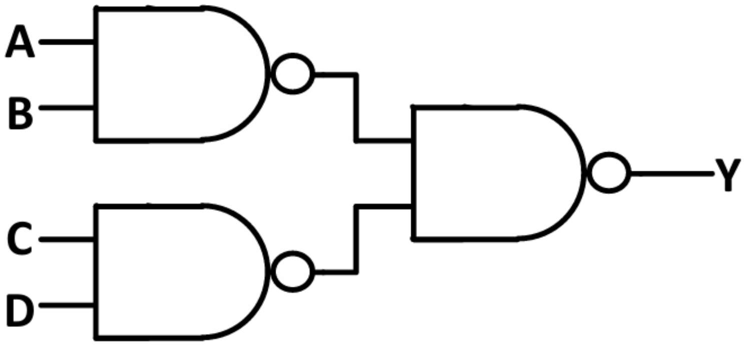
COMET FWC 071

EE 2018 Q-14

Q.In the logic circuit shown in the figure, Y is given by



1. A) $Y = ABCD$
2. B) $Y = (A + B)(C + D)$
3. C) $Y = A + B + C + D$
4. D) $Y = AB + CD$

Step 1: First Two Gates

The upper and lower gates are NAND gates.

$$X = (AB)' \tag{1}$$

$$W = (CD)' \tag{2}$$

Step 2: Final Gate

The outputs of the two NAND gates are fed to another NAND gate.

$$Y = (XW)' \tag{3}$$

Substituting X and W,

$$Y = [(AB)'(CD)']' \tag{4}$$

Step 3: Apply De Morgan's Theorem

$$(A'B')' = A + B \tag{5}$$

More generally,

$$(P'Q')' = P + Q \tag{6}$$

Let

$$P = AB \tag{7}$$

and

$$Q = CD \tag{8}$$

Therefore,

$$Y = (AB) + (CD) \tag{9}$$

Hence,

$$Y = AB + CD \tag{10}$$

Correct Option

$$\text{(D) } Y = AB + CD \tag{11}$$